LongCat-Next: Lexicalizing Modalities as Discrete Tokens

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Research Questions

- How can non-linguistic modalities (vision, audio) be effectively represented within a discrete token space comparable to language tokens?
- Can discrete visual modeling achieve performance parity with continuous representations for both understanding and generation tasks?
- How can understanding and generation objectives be unified under a single autoregressive framework without performance degradation?

Core Idea

Current multimodal systems treat vision and audio as external attachments to language models, leading to fragmented architectures. This paper introduces DiNA (Discrete Native Autoregressive), which represents all modalities as discrete tokens in a shared space, enabling unified next-token prediction across text, vision, and audio. The key insight is using Semantic-and-Aligned Encoders (SAE) with Residual Vector Quantization (RVQ) to achieve "semantic completeness" - preserving sufficient information for both understanding and generation. The approach leverages residual connections in encoders that inherently preserve low-level information pathways, enabling high-fidelity reconstruction even without explicit supervision.

Key Formula

Semantic Completeness Condition

$$P(A|z, Q) \approx P(A|I, Q)$$

A represents optimal response, z is discrete token sequence, I is original image, Q is image-centric query. This formalizes that discrete representation z must preserve sufficient information to approximate the posterior distribution of the original continuous signal I for any task, ensuring both discriminative invariance and generative sufficiency.

Residual Vector Quantization

$$r_0 = f_{proj}(z), \quad \hat{q}_l = VQ(r_{l-1}), \quad r_l = r_{l-1} - \hat{q}_l, \quad \hat{z} = \sum_{l=1}^{L} \hat{q}_l$$

r_0 is projected latent representation, \hat{q}_l is quantized feature at level l, r_l is residual error, L is number of codebook levels. This hierarchical quantization preserves information by recursively encoding residuals, enabling both semantic understanding and fine-grained generation from the same discrete tokens.

Key Figure

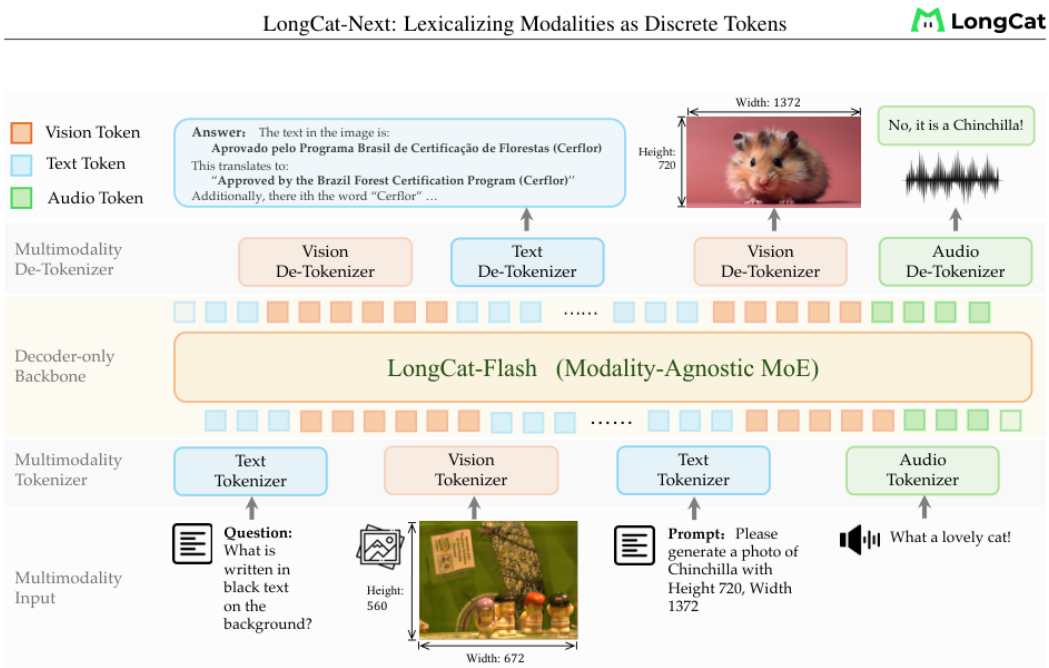

Figure 2: Overview of the LongCat-Next architecture, designed under a Discrete Native Autoregression (DiNA) paradigm that extends multimodality into a native language-style modeling framework via paired tokenizers.

Figure 2: Shows the complete DiNA architecture with modality-specific tokenizer-detokenizer pairs feeding into a unified decoder-only backbone, illustrating how text, vision, and audio are processed under the same autoregressive framework.

Limitations

- Discrete representations still show initial performance gaps compared to continuous models, requiring more training data to converge
- High-resolution visual processing remains computationally expensive despite compression ratios
- Current data mixture contains limited high-resolution samples, affecting GUI and fine-grained visual tasks
- The unified training paradigm requires careful balancing between modality-specific data to prevent degradation